

ATHENA: An Image Captioning Pipeline for Constrained Underwater Environments

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Rodrigo Emmanuel D. Roy

Krishna Anne E. Chan

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Prof. Prospero C. Naval, Jr., Ph.D.
Thesis Adviser

Asst. Prof. Carlo R. Raquel
Thesis Adviser

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Abstract

Autonomous underwater vehicles (AUVs) are constrained by low-bandwidth acoustic communication channels and limited onboard computation. Current approaches either rely on cloud processing or direct image transmission, both of which are infeasible for untethered AUV operation. In this work, we introduce ATHENA: AI Transformer for Hydro-Environmental Navigation and Analysis, an edge-capable Vision-Language Model (VLM) pipeline that enables AUVs to describe underwater scenes as natural language, supporting efficient acoustic transmission and real-time situational awareness. To address domain-specific visual degradation such as color distortion, light attenuation, and haze, ATHENA applies a multi-step image enhancement pipeline before VLM inference. Through 4-bit NF4 quantization, ATHENA operates within a 653 MB memory footprint, enabling deployment on embedded hardware such as the NVIDIA Jetson Orin series. ATHENA outperforms all prior specialized underwater captioning models on the standard benchmark, and further evaluation on UW Bench reveals that the enhancement pipeline is conditionally critical: it produces consistent improvements on visually degraded inputs while being unnecessary for clearer images. Despite using only 1.37 billion parameters, ATHENA surpasses 13 of 14 frontier vision-language models in zero-shot evaluation on UW Bench, including GPT-4o and Gemini-2.5-Flash, with particularly strong performance on fluency-based metrics.

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Chapter 1

Introduction

Autonomous underwater vehicles (AUVs) are increasingly deployed for marine exploration, environmental monitoring, and subsea infrastructure inspection. Long-duration autonomous missions require onboard perception systems that can summarize visual scenes and transmit only essential information to remote operators. However, this is constrained by two defining characteristics of underwater robotics: severely limited communication bandwidth and restricted onboard computational resources [3]. Underwater communication is typically achieved via acoustic channels, which use sound waves to transmit information over long distances, since radio signals attenuate rapidly in water and optical signals require line-of-sight. Acoustic channels have low bandwidth, high latency, and multipath interference, making raw or high-resolution image transmission and cloud-based processing impractical, especially for untethered AUVs [4][5][6]. Many state-of-the-art vision models also exceed the compute and memory budgets of embedded AUV hardware [5][7].

Underwater imagery further complicates perception due to light attenuation, scattering, haze, and color distortion, which vary with depth, turbidity, and lighting conditions. Prior work highlights the need for enhancement and restoration pipelines to

address these challenges before downstream processing [5].

To overcome these limitations, recent approaches focus on conveying semantic content rather than raw images. Textual descriptions offer a compact, bandwidth-efficient alternative [4][6]. Vision-language models (VLMs) such as those deployed in SINGER [8] and SmolVLM [7] demonstrate the feasibility of onboard semantic understanding, though existing architectures are often too large for edge deployment or are not tailored to underwater imagery.

Inspired by these works, we propose ATHENA, an edge-capable underwater scene captioning system that enhances raw images and produces concise textual descriptions suitable for low-bandwidth transmission, enabling AUVs to communicate semantically meaningful information in real time.

The remainder of this paper is organized as follows: Chapter 2 reviews related work on vision-language models, semantic communication, and underwater image enhancement. Chapter 3 presents the ATHENA framework, including the dataset, methodology, and experimental setup. Chapter 4 details experiments and analyzes results. Finally, Chapter 5 concludes the paper.

1.1 Statement of the Problem

Existing underwater image captioning approaches face three interconnected challenges. First, most state-of-the-art VLMs are pre-trained on terrestrial image datasets and consistently underperform on underwater scenes due to the significant domain gap caused by color attenuation, backscatter, and turbidity [1]. Second, underwater imagery is inherently degraded by haze, low contrast, and scattering, and current enhancement methods lack consistency across diverse underwater conditions [9]. Third,

AUVs operate under strict onboard computational and memory budgets and rely on low-bandwidth acoustic communication channels, making cloud-based processing and raw image transmission infeasible for real-time operation [3].

1.2 Significance of this Study

This study addresses the need for a practical, edge-deployable system that enables AUVs to communicate the semantic content of underwater scenes without relying on high-bandwidth channels or cloud processing. By introducing ATHENA, an edge-capable VLM pipeline with domain-adaptive fine-tuning and a tailored image enhancement pipeline, this work advances the state of the art in underwater image captioning and demonstrates that compact, quantized VLMs can achieve competitive performance on standard benchmarks while meeting the resource constraints of embedded AUV hardware. The findings contribute to the broader effort of enabling long-duration autonomous underwater missions with onboard, real-time scene understanding.

1.3 Scope of this Study

This study focuses on image captioning for underwater scenes using two datasets: the Li et al. dataset of 3,176 underwater images with five human-annotated captions each [1], and UW Bench [2], a large-scale benchmark of 15,003 high-resolution images with expert-verified captions covering 158 object categories. Three VLM backbones are evaluated: SmolVLM-Instruct, BLIP-2 OPT 2.7B, and Qwen3.5-VL, under a systematic ablation across LoRA ranks and DoRA configurations. The image enhancement pipeline is based on a seven-step protocol adapted from Wang et al. [9],

with CLAHE substituted for the original contrast stretching step. Deployment feasibility is assessed under 4-bit NF4 quantization targeting embedded edge hardware. This work does not address real-time physical deployment on AUV hardware, underwater object detection, video captioning, or the design of acoustic communication protocols.

Chapter 2

Review of Related Literature

Recent advances in autonomous robotics and computer vision have explored various approaches for perception, communication, and image enhancement. In the context of underwater scene understanding, three main areas are particularly relevant: vision-language models (VLMs) for robotic perception, semantic communication methods for low-bandwidth environments, and specialized techniques for underwater image enhancement.

2.1 Communication-Constrained Multi-AUV Cooperative SLAM [3]

Multi-AUV cooperative SLAM enables collaborative localization through direct encounters (relative range measurements) and indirect encounters (shared observations of seabed features). However, underwater acoustic channels operate at approximately 1,500 m/s propagation speed with throughputs of only 10–100 bytes per second, 20–50% packet dropout, and TDMA constraints, making raw sensor data transmission

impractical. A graph-based C-SLAM algorithm addresses this by locally marginalizing vehicle pose estimates via the Schur complement, condensing accumulated observations into compact virtual measurements. A sparsification step based on constrained convex optimization further reduces packet size by minimizing the Kullback-Leibler divergence between the original and approximate distributions while enforcing a predetermined sparse Jacobian structure. The resulting payload scales linearly with observed features and remains constant in time even under dropped packets. Simulation with two AUVs demonstrates bounded localization error at 50% dropout with a maximum packet size of 192 bytes per 10-second interval.

These severe bandwidth constraints directly motivate ATHENA’s semantic captioning approach: if even compressed SLAM packets must remain under 200 bytes to be acoustically viable, transmitting compact natural language scene descriptions is far more practical than any form of image transmission.

2.2 Robotics Perception and Control: Key Technologies and Applications [10]

This comprehensive review surveys nine sensor categories used in robotic control systems, including IMUs, visual sensors, LiDAR, tactile sensors, and EMG sensors, categorized as either proprioceptive (measuring internal robot states) or exteroceptive (gathering information from the external environment). Vision sensors occupy a central role, supporting object recognition, visual SLAM, autonomous navigation, medical surgery, and deep learning-based robotic grasping across industrial, medical, and exploratory domains.

The survey situates VLMs as an emerging class of exteroceptive perception module

that converts visual input into structured semantic representations for downstream tasks such as navigation and scene understanding. ATHENA adopts this paradigm by deploying a VLM as an onboard sensor module that produces textual descriptions suitable for transmission over low-bandwidth acoustic channels, analogous to how onboard ATR systems convert raw sonar returns into compact feature representations.

2.3 Real-Time Vision–Language Analysis for Autonomous Underwater Drones [5]

A cloud-edge framework for real-time VLM analysis in AUVs uses Qwen2.5-VL, a 72-billion-parameter model, by offloading inference to a cloud server while retaining lightweight preprocessing on the edge device. The edge performs frame sampling at 1 fps and a three-stage image enhancement pipeline (color correction, backscatter reduction, white balance), while communication relies on a 5 Mbps fiber-optic link with adaptive JPEG quality scaling and bounded frame buffering. Simulation across pipeline inspection, coral reef monitoring, and wreck investigation scenarios demonstrates consistent scene comprehension with end-to-end latencies near 1 second.

However, the reliance on cloud infrastructure and tethered optical fiber makes this approach unsuitable for untethered AUV operations, underscoring the need for a fully onboard, lightweight inference pipeline such as the one adopted in ATHENA.

2.4 SINGER: An Onboard Generalist Vision-Language Navigation Policy for Drones [8]

SINGER addresses open-vocabulary drone navigation by integrating a photorealistic language-embedded 3D Gaussian Splatting simulator, an RRT*-based trajectory

generator, and a lightweight visuomotor policy trained on synthetic data. The system uses CLIPSeg to convert RGB images into semantic heatmaps as policy inputs, enabling generalization across unseen environments. To achieve real-time inference on the NVIDIA Jetson Orin Nano 8GB, CLIPSeg is traced to ONNX and executed via the CUDA runtime at 12 Hz. Hardware experiments demonstrate 76.67% goal proximity success across 30 trials, outperforming a classical PD controller baseline by 23.33% with 10% fewer collisions.

This result establishes a key precedent for ATHENA’s edge deployment strategy, confirming that VLM-based semantic understanding can run in real time on the Jetson Orin Nano without cloud offloading and validating the hardware feasibility of onboard inference in resource-constrained robotic platforms.

2.5 SmolVLM: Redefining Small and Efficient Multimodal Models [7]

SmolVLM is a family of compact multimodal models that systematically investigates architectural configurations, tokenization strategies, and data composition to maximize VLM performance at minimal memory footprints. The architecture pairs a SigLIP vision encoder with a SmolLM2 language backbone and uses pixel shuffle compression to reduce visual tokens by a factor of 16. Three variants are produced: SmolVLM-256M (< 1 GB RAM), SmolVLM-500M (1.2 GB), and SmolVLM-2.2B (4.9 GB). The 2.2B model achieves 59.8% average score across nine benchmarks using only 4.9 GB VRAM, outperforming Qwen2VL-2B which requires 13.7 GB. SmolVLM-256M, operating under 1 GB RAM, significantly surpasses the 80B Idefics model despite a 300-fold parameter gap.

SmolVLM-Instruct serves as the lightweight backbone evaluated in ATHENA, representing the sub-1 GB end of the model scale spectrum and confirming that competitive captioning performance is achievable within the memory budgets of embedded AUV hardware.

2.6 BLIP-2: Bootstrapping Language-Image Pre-Training with Frozen Image Encoders and Large Language Models [11]

BLIP-2 proposes a compute-efficient pre-training strategy that bootstraps vision-language learning from frozen image encoders and frozen LLMs using a lightweight Querying Transformer (Q-Former) with only 188 million trainable parameters, 54 times fewer than Flamingo80B, while surpassing it by 8.7% on zero-shot VQAv2. Q-Former employs 32 learned query vectors to extract task-relevant visual information through cross-attention, serving as an information bottleneck between the vision and language components. The model is pre-trained in two stages: first learning visual representations through contrastive, generative, and matching objectives on a 129M-image dataset; then projecting Q-Former outputs as soft visual prompts to a frozen LLM for language generation. On zero-shot captioning, BLIP-2 achieves a CIDEr score of 121.6 on the NoCaps validation set.

However, systematic analyses identify BLIP-2 as producing formulaic outputs with low lexical diversity and limited reasoning consistency [12]. In ATHENA, BLIP-2 OPT 2.7B serves as the intermediate-scale backbone in the ablation study, providing a reference point between SmolVLM and Qwen3.5-VL.

2.7 Toward Human-Level Understanding: A Systematic Review of Vision-Language Models for Image Captioning [12]

A systematic review evaluates eight VLMs for image captioning under zero-shot conditions on MS-COCO, combining automatic metrics (CLIPScore, perplexity, lexical diversity) with human evaluation across 1,000 captions per model. LLaVA achieves the highest CLIPScore (0.757), while BLIP-2 exhibits the lowest lexical diversity and highest perplexity (97.1), characteristic of its conservative Q-Former bottleneck. A structured WH-question analysis across six semantic dimensions (Subject Detail, Action Detail, Location Detail, Time Detail, Manner Detail, and Purpose/Reasoning) reveals a clear progression from early models producing basic subject-action-location statements to advanced models generating multi-component narratives approaching human-level descriptive richness.

Critically, the review identifies that VLMs pre-trained on terrestrial datasets consistently struggle with domain-shifted inputs, exhibiting degraded performance on imagery with fundamentally different visual statistics. This finding directly motivates ATHENA’s fine-tuning strategy: standard zero-shot VLMs cannot produce accurate underwater scene descriptions without supervised adaptation on domain-specific data.

2.8 TensorRT-LLM: Embedded Inference for Large Vision-Language Models [13]

The TensorRT Edge-LLM framework provides tooling for deploying large language and vision-language models on embedded hardware through aggressive quantization and inference optimization. Targeting devices such as the NVIDIA Jetson family, the

framework applies INT4 and INT8 weight quantization, KV-cache compression, and attention kernel fusion to reduce memory footprint and inference latency. Its 4-bit weight quantization achieves approximately $4\times$ memory reduction compared to FP16 inference, making large VLMs viable on platforms with limited unified memory.

Support for Qwen3.5-VL through this framework justifies its selection as a viable ATHENA backbone despite its larger parameter count, with 4-bit NF4 quantization via the *bitsandbytes* library confirming deployment feasibility on the Jetson Orin hardware platform.

2.9 Underwater Image Captioning: Challenges, Models, and Datasets [1]

Underwater image captioning is hindered by three challenges: models pre-trained on terrestrial images create a systematic domain gap that causes misidentification of marine objects; underwater images exhibit degraded quality from wavelength-dependent absorption, backscatter, and turbidity; and existing underwater datasets either lack captions or are too small for training. The proposed UICM-SOFF model addresses these by using a physics-based submergence GAN to generate 80,000 degraded training images, pre-training a Meta-ResNeXtor feature extractor via meta-learning, and fusing scene-level features with YOLOv8 object-level features through a cross-attention Transformer encoder. The constructed dataset comprises 3,176 images across ten categories (predominantly fish, coral reefs, and divers) with five human-written captions each, totaling 15,880 captions.

This dataset serves as the primary training and evaluation benchmark for ATHENA, and UICM-SOFF’s scores (BLEU-1: 0.7249, BLEU-4: 0.3953, ROUGE-L: 0.5888,

CIDEr: 1.2831) define the state-of-the-art baseline against which ATHENA’s results are directly compared.

2.10 Semantic Communication Based on Large Language Model for Underwater Image Transmission [4]

An LLM-based semantic communication framework for underwater image transmission represents image content as compact semantic information rather than pixel data, reducing transmitted data to approximately 0.8% of the original image size. The semantic encoder passes the image to a visual LLM to generate text encoding high-level content, identifies key image regions via bounding box prediction at reduced resolution, and aggressively resamples the remaining background; the combined payload is approximately 27,525 bytes versus the original 3,145,728 bytes. At the receiver, a diffusion model reconstructs the full image from three guidance signals conditioned on the recovered text and key regions. Evaluation on the Semantic Segmentation of Underwater Imagery (SUIM) dataset shows the highest CLIP scores across all tested Signal-to-Noise Ratio (SNR) levels (0–18 dB).

Both this framework and ATHENA share the core insight that language-based representations serve as an effective acoustic communication payload to overcome the fundamental bandwidth limitations of underwater channels.

2.11 An Underwater Acoustic Semantic Communication Approach to Underwater Image Transmission [6]

This work proposed an acoustic semantic communication framework that extracts semantic features from underwater images using deep neural networks and encodes them for delivery over bandwidth-limited, high-latency acoustic channels. Semantic encoding separates task-relevant content from perceptually redundant detail, achieving significant bandwidth savings while maintaining downstream task performance under challenging acoustic conditions.

These findings reinforce the practical viability of semantic representations for AUV communication and provide quantitative support for ATHENA’s design choice to transmit VLM-generated captions rather than compressed images over acoustic channels.

2.12 Semantic Communications: Overview, Open Issues, and Future Research Directions [14]

A broad survey of semantic communication systems identifies three communication levels: technical (reliable bit transmission), semantic (accurate meaning transfer), and effectiveness (achieving desired receiver effects). Semantic systems encode only semantically relevant content while discarding redundant data, which is particularly valuable in bandwidth-constrained environments such as underwater acoustic channels where available bitrates are orders of magnitude lower than those needed for raw image transmission.

This framework underpins the rationale for ATHENA’s use of VLM-generated

captions as the communication payload: by transmitting meaning rather than pixel representations, ATHENA operates as a semantic communication system at the second level, achieving efficiency gains through language-level scene representation.

2.13 From Semantic Communication to Semantic-Aware Networking: Model, Architecture, and Open Problems [15]

Semantic-aware networking decouples semantic encoding from physical-layer transmission, enabling network resources to be allocated based on semantic importance rather than treating all bits uniformly. The proposed architecture introduces a semantic source encoder, a channel encoder, and a semantic decoder that reconstructs intended meaning without necessarily recovering original data exactly. Unlike conventional compression, which minimizes bitrate subject to distortion constraints, semantic encoding minimizes the information needed to achieve a specified level of semantic fidelity.

This decoupled architecture supports ATHENA’s goal of generating compact captions that can be prioritized for transmission over low-bandwidth acoustic channels, where operator situational awareness depends on scene content rather than image fidelity.

2.14 Deep Learning Enabled Semantic Communication Systems [16]

End-to-end deep learning methods can jointly optimize semantic encoding and channel coding, outperforming conventional bit-based communication under low SNR conditions. The system trains a semantic encoder-decoder pair with the channel as a non-trainable intermediate, learning which input aspects are most important for downstream tasks and encoding them more robustly. Experiments show significant improvements in semantic fidelity at low SNR values typical of underwater acoustic channels.

This lends support to ATHENA’s use of text captions as a robust underwater transmission format, since natural language descriptions constitute a compact semantic representation that preserves task-relevant content more reliably than compressed image data under noisy channel conditions.

2.15 Deep Learning-Based Image Semantic Coding for Semantic Communications [17]

A deep learning-based framework transmits semantic image representations rather than compressed pixel data through joint source-channel coding. The semantic encoder learns scene-level abstractions appropriate for downstream tasks while discarding irrelevant texture detail, and the decoder uses generative modeling to reconstruct images from the compact semantic code. The approach scales to very low bitrate regimes corresponding to the available bandwidth of underwater acoustic channels.

While this framework transmits learned continuous-valued semantic features, ATHENA extends the same principle to discrete natural language representations: VLM-generated

captions encode the semantically relevant aspects of underwater scenes in a format that is both human-interpretable and efficiently transmissible over acoustic channels.

2.16 Semantic Segmentation-Based Semantic Communication System for Image Transmission [18]

This system applies semantic segmentation as the transmission representation for image communication, extracting pixel-level semantic labels instead of raw pixels. Segmentation maps require significantly lower bitrates while preserving object category distribution and scene structure, sufficient for many monitoring and inspection tasks. The system demonstrates robust performance under channel noise conditions typical of wireless and acoustic links.

This approach contextualizes ATHENA’s choice of natural language captions as a compact alternative: while segmentation maps encode spatial structure, captions encode semantic content in a directly human-readable form that can be interpreted without specialized decoding equipment, making them well-suited for transmission to surface operators.

2.17 Sea-thru: A Method For Removing Water From Underwater Images [19]

Underwater images suffer from systematic color distortion and contrast loss caused by wavelength-dependent absorption and scattering. Sea-thru addresses these by proposing a physics-based image formation model that separates signal attenuation from backscatter as functions of range-to-scene depth. By estimating and subtracting the backscatter contribution, then compensating for channel-specific attenuation, the

method recovers true object reflectances validated against photographs taken in air.

However, the requirement for depth information, typically obtained from multi-image structure-from-motion reconstruction, limits real-time onboard applicability. The method also does not address contrast or sharpness degradation beyond color correction. These limitations motivate the more comprehensive multi-step enhancement pipeline adopted in ATHENA, which addresses color, contrast, and sharpness without requiring depth information.

2.18 Meta Underwater Camera: A Smart Protocol for Underwater Image Enhancement [9]

Underwater images suffer from three categories of degradation: color deviations from wavelength-dependent attenuation, low contrast from light scattering, and detail blur from backscatter. The Meta Underwater Camera (MUC) protocol addresses all three through a cascaded seven-step pipeline: (1) attenuated channel compensation, (2) white balance via YCbCr mean method, (3) tone mapping, (4) HSL saturation adjustment, (5) contrast stretching, (6) gamma correction, and (7) high-pass fusion. A reinforcement learning agent trained with PPO dynamically configures the eight pipeline parameters using a 512-dimensional multi-color-space feature vector and a reward based on UCIQE and UIQM quality metrics. Comparative evaluation against nine methods on the UIEB dataset demonstrates superior image quality across diverse scene types.

This seven-step protocol forms the backbone of ATHENA’s preprocessing module, with one modification: the contrast stretching step is replaced with CLAHE for improved local contrast preservation in heterogeneous underwater scenes.

2.19 Deep Learning Based Deep-Sea Automatic Image Enhancement and Animal Species Classification [20]

A combined image enhancement and classification pipeline targets deep-sea benthic megafauna from footage captured at 870 meters depth. The enhancement proceeds in two phases: classical preprocessing (white balance, gamma correction, CLAHE) followed by a 14-layer residual CNN that generates enhanced images. The full pipeline achieves the highest image quality scores (UIQM of 2.585, UCIQE of 2.406), and classification accuracy on CNN-enhanced images reaches 79.44% compared to 66.44% on unenhanced data.

The substantial improvement in downstream task accuracy confirms that pre-processing enhancement is a necessary precondition for effective visual understanding, directly supporting the inclusion of a dedicated image enhancement pipeline in ATHENA’s architecture.

2.20 Underwater Image Enhancement Method Based on Entropy Weight Fusion [21]

An entropy weight fusion framework constructs three complementary enhancement branches: gray-world white balance with adaptive gamma correction, CLAHE with bilateral filtering, and Single-Scale Retinex with guided filtering in HSV space. The three outputs are fused using information-theoretic weights derived from UIQM, entropy, and average gradient, requiring no reference images. Evaluation on the UIEBD dataset demonstrates superior performance over competing methods across greenish, bluish, hazy, and limited-illumination scenes.

These results reinforce ATHENA’s multi-step enhancement design: no single technique can address the full spectrum of underwater degradation, and combining complementary approaches achieves robust improvement across the diverse conditions encountered in real AUV deployments.

2.21 Comparative Analysis of Underwater Image Enhancement Methods in Different Color Spaces [22]

A comparative analysis evaluates standard contrast stretching and CLAHE as representative global and local contrast enhancement techniques for underwater images. CLAHE consistently outperforms contrast stretching by operating on local tiles independently, enhancing dark regions without over-brightening well-illuminated areas, while its contrast limit parameter prevents excessive noise amplification.

These findings justify the substitution of CLAHE for the original contrast stretching step in ATHENA’s preprocessing pipeline, as its local adaptivity is particularly beneficial for underwater images that frequently contain both dark shadowed regions and brightly illuminated foreground subjects.

2.22 UW Bench: A Comprehensive Vision-Language Benchmark for Underwater Understanding [2]

UWBench is the first large-scale vision-language benchmark designed specifically for comprehensive underwater scene understanding. It comprises 15,003 high-resolution underwater images spanning shallow reefs, mid-depth zones, and deep-sea habitats, covering 158 object categories. Each image is annotated with a detailed caption, object referring expressions, and visual question-answer pairs, all produced through

a semi-automatic pipeline combining GPT-5-assisted generation with rigorous three-stage verification by marine biology experts. The benchmark establishes three evaluation tasks: UWBench-Cap for detailed image captioning, UWBench-Ref for visual grounding of marine organisms, and UWBench-VQA for multimodal underwater reasoning.

Experiments evaluating 14 state-of-the-art VLMs on UWBench-Cap, including GPT-4o, GPT-5, Gemini-2.5-Flash, and several open-source models, show that all models exhibit substantial performance drops compared to terrestrial benchmarks, confirming that underwater vision-language understanding remains a challenging open problem. UWBench provides the extended evaluation environment for ATHENA’s preprocessing ablation and frontier model comparison, offering a harder and more ecologically diverse benchmark than the training dataset.

2.23 LoRA: Low-Rank Adaptation of Large Language Models [23]

Low-Rank Adaptation (LoRA) addresses the challenge of fine-tuning large pretrained models by freezing the original weights and injecting trainable low-rank decomposition matrices into each attention layer. For a weight matrix $W \in \mathbb{R}^{d \times k}$, LoRA parameterizes the update as $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$. This reduces the number of trainable parameters from $d \times k$ to $r(d+k)$, enabling efficient fine-tuning with substantially lower memory overhead. On GPT-3 175B, LoRA with rank 4 matches full fine-tuning performance on several downstream tasks while reducing the trainable parameter count by a factor of 10,000 and GPU memory requirements by two thirds.

LoRA is the primary fine-tuning strategy used in ATHENA’s backbone adaptation, evaluated across ranks $r \in \{8, 16, 32\}$ for all three backbone models to identify the optimal capacity-efficiency trade-off for the underwater captioning domain.

2.24 DoRA: Weight-Decomposed Low-Rank Adaptation [24]

Weight-Decomposed Low-Rank Adaptation (DoRA) extends LoRA by decomposing each pretrained weight matrix into a magnitude component and a directional component, then applying LoRA only to the directional part while learning the magnitude separately. This decomposition draws from weight normalization and allows DoRA to more closely replicate the learning behavior of full fine-tuning: analysis shows that full fine-tuning consistently adjusts magnitude and direction jointly, while standard LoRA tends to over-update direction and under-adjust magnitude. DoRA eliminates this gap without adding inference overhead, since the decomposed form collapses back to a standard weight matrix at test time. On commonsense reasoning, visual instruction tuning, and image-text alignment tasks, DoRA consistently outperforms LoRA at the same rank by up to 4.4% with no additional latency cost.

In ATHENA, DoRA is evaluated alongside standard LoRA across all backbone and rank configurations. The ablation results show DoRA consistently improves scores, with the largest gains on Qwen3.5-VL where stronger pretrained representations benefit most from the finer-grained directional and magnitude decomposition.

Chapter 3

Methodology

This chapter describes the data, methodology, and experimental setup used to develop and evaluate ATHENA. We first present the underwater image captioning datasets used for training and evaluation. We then describe the overall system architecture, including the image enhancement pipeline and VLM backbones. Finally, we detail the ablation study design, fine-tuning configurations, and evaluation metrics used to assess model performance.

3.1 Data

Underwater Image Captioning Dataset (Li et al.)

For our study, we used a publicly available underwater image captioning dataset collected by Li et al. [1], which consists of 3,176 high-quality underwater images. We selected this dataset because it is the most comprehensive publicly available underwater dataset with multiple human-written captions per image. Other popular underwater datasets, such as UIEB [25], SQUID [26], UID2021 [27], URPC2021 [28], and DUO [29], either lack captions, focus solely on object detection, or are much smaller in scale.

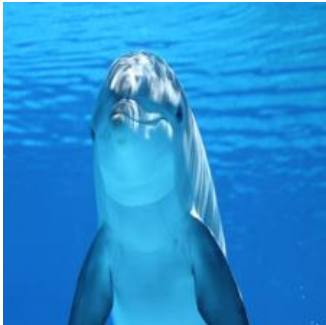
The images cover a wide range of resolutions and were resized to 224×224 pixels for consistency and ease of use. The dataset includes diverse underwater scenes, such as small distant targets, dense groups, visual blur, and low-light conditions, representing real-world underwater environments. The dataset also spans ten categories, predominantly fish (1861), coral reefs (1587), and divers (850).

Each image was annotated with five distinct captions provided by volunteers with domain expertise, resulting in a total of 15,880 captions. These captions, created as part of the original research, describe object morphology, color, quantity, motion, and additional scene context, ensuring rich vocabulary and syntactic diversity. Table 3.1, reproduced from Li et al. [1], presents some representative examples from the dataset. The dataset provides a comprehensive benchmark for underwater scene understanding and forms the foundation for training and evaluating our vision-language model.

UWBench: Extended Evaluation Dataset

To evaluate ATHENA under more challenging and realistic underwater conditions, we additionally use UWBench [2], a recently introduced large-scale vision-language benchmark specifically designed for comprehensive underwater scene understanding. UWBench comprises 15,003 high-resolution underwater images spanning diverse aquatic environments (including shallow reefs, mid-depth zones, and deep-sea habitats) covering 158 object categories. Unlike the Li et al. dataset, which consists of relatively clear images captured under controlled or favorable conditions, UWBench images exhibit richer variation in water clarity, illumination, occlusion, turbidity, and color distortion, more closely reflecting the degraded visual conditions encountered in real-world constrained underwater deployments. Each image in UWBench

Table 3.1: Sample images and captions from the Li et al. dataset [1]

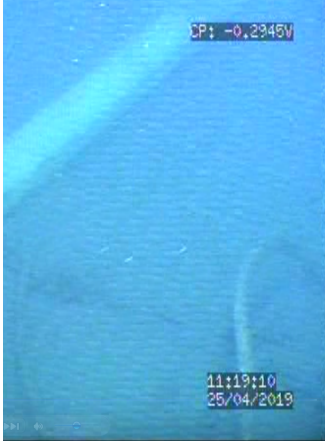
Image	Ground truth captions
	GT-1: Coral reefs in seawater come in a variety of colors, such as dark-brown, light-brown, and white. GT-2: An underwater coral reef full of life, with all kinds of corals growing freely. GT-3: Coral reefs in the water grow corals of various shapes and colors. GT-4: An undersea coral reef with corals of various colors and shapes. GT-5: There are many corals of various shapes and colors growing on the underwater reefs.
	GT-1: A gray dolphin swims in the blue seawater. GT-2: There is a gray dolphin swimming at the bottom of sea. GT-3: A close-up of a dolphin swimming in the blue seawater. GT-4: There is a gray dolphin swimming in the seawater. GT-5: A gray dolphin swims at the bottom of sea.

is annotated with comprehensive human-verified captions produced through a rigorous semi-automatic pipeline combining GPT-5 assisted generation with three-stage marine biology expert review, ensuring scientific accuracy and ecological validity.

We use the image captioning split of UWBench (UWBench-Cap) for two purposes: to evaluate the effect of the image enhancement pipeline on degraded inputs, and to compare ATHENA against frontier vision-language models in a common zero-shot benchmark. The contrast between the two datasets allows us to assess whether the preprocessing pipeline is universally beneficial or conditionally necessary depending on the visual quality of the input images. Table 3.2 presents representative examples from the dataset. A detailed account of the UWBench construction pipeline, including

image collection, attribute extraction, GPT-5 assisted annotation, and the three-stage marine biology expert verification process, is provided in Appendix B.2.

Table 3.2: Sample images and captions from the UWBench dataset [2]

Image	Ground truth caption
	GT: A blue-tinted underwater scene shows a rocky seabed with uneven reef structures and scattered hard surfaces. In the middle of the frame, a single large mantaray forms a dark, diamond-shaped silhouette against the lighter bottom. The wings extend broadly left and right, with the body covering much of the lower half of the image. Water clarity appears moderate, giving the scene a slightly hazy look.
	GT: An underwater view shows a sandy seabed under somewhat turbid, blue water. A single large submarine pipeline runs diagonally from the upper-left toward the center, occupying much of the upper half of the frame with a smooth, cylindrical appearance. The surrounding seafloor appears flat and fine-grained, with no obvious coral or rocks. Overall, the scene is a sandy bottom dominated by one prominent pipeline.

3.2 Methodology

System Architecture

The ATHENA pipeline begins with image capture, sending the raw image and a system prompt as input. The image first passes through a multi-step enhancement pipeline, after which the enhanced image is fed into the VLM, which generates a structured textual description. The output text is then transmitted over a low-bandwidth acoustic channel to a remote operator. The overall architecture is illustrated in Figure 3.1.

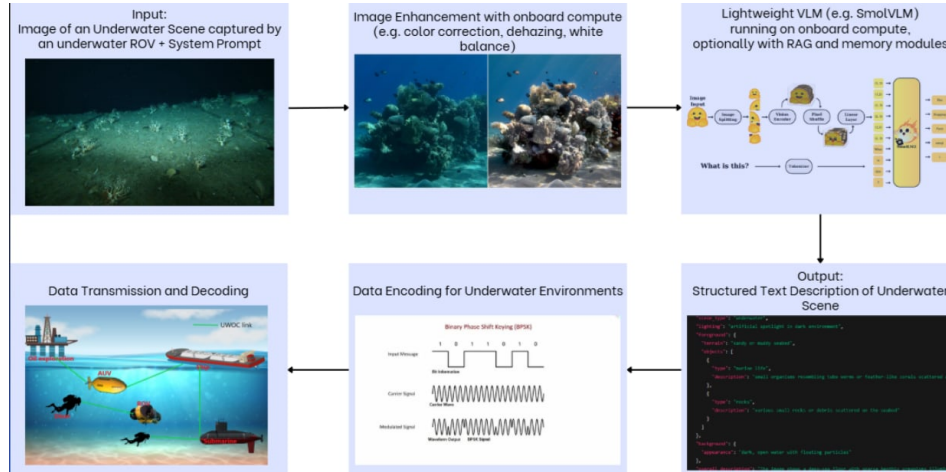


Figure 3.1: ATHENA system architecture

Image Enhancement Pipeline

Raw underwater images are enhanced before VLM inference to improve feature extraction and caption accuracy. The pipeline is based on the seven-step protocol proposed by Wang et al. [9], with one modification: the original contrast stretching step is replaced with CLAHE for better local contrast enhancement [22]. The seven steps are: (1) **Attenuated Channel Compensation (ACC)**, which restores

absorbed reds and yellows; (2) **White Balance** (YCbCr mean method), which corrects blue/green color casts; (3) **Tone Mapping**, which enhances luminance across dark and bright regions; (4) **HSL Saturation Adjustment**, which boosts color vividness; (5) **CLAHE**, which improves local contrast without amplifying noise; (6) **Gamma Correction**, which adjusts brightness for under- and overexposed regions; and (7) **High-Pass Fusion**, which sharpens edges and fine details for better VLM feature recognition. These steps address three main visual issues: the first four correct color distortions, the next two enhance contrast, and the final step sharpens image details. The overall process is illustrated in Figure 3.2.

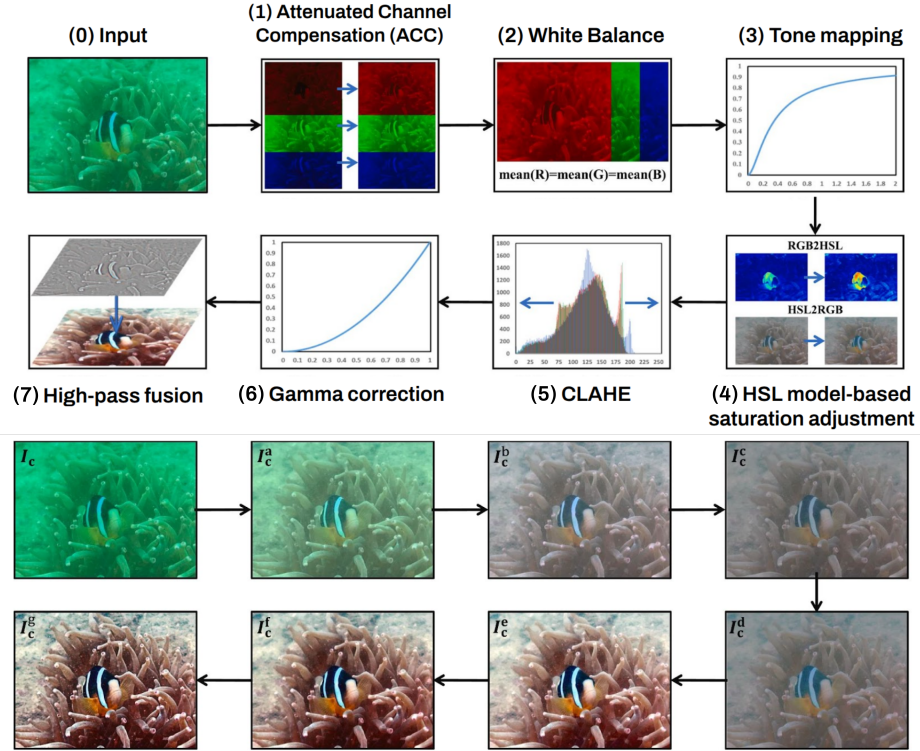


Figure 3.2: Image enhancement pipeline (top) and sample output (bottom)

VLM Backbone Selection

Three VLM backbones of varying architectural scales are evaluated to identify the most suitable model for underwater scene captioning under resource constraints. **SmolVLM-Instruct** [7] represents the lightweight end, offering a sub-1 GB memory footprint suitable for real-time onboard inference. **BLIP-2 OPT 2.7B** [11] serves as an intermediate-scale reference with strong zero-shot captioning potential but known limitations in linguistic consistency [12]. **Qwen3.5** [13] represents the large-scale upper bound, with stronger scene understanding capabilities and support for embedded inference via NVIDIA TensorRT Edge-LLM. Parameter-efficient fine-tuning is applied to all three backbones using Low-Rank Adaptation (LoRA) [23] and Weight-Decomposed Low-Rank Adaptation (DoRA) [24]. DoRA extends LoRA by decomposing pretrained weights into magnitude and directional components, more closely resembling full fine-tuning behavior without additional inference overhead [24]. Each backbone is evaluated across three LoRA ranks ($r = 8, 16, 32$) with DoRA enabled and disabled, yielding six configurations per backbone and 18 total, as summarized in Table 3.3.

Table 3.3: Ablation study configurations

VLM Backbone	LoRA Rank	DoRA	Configs
SmolVLM-Instruct	8, 16, 32	True / False	6
BLIP-2 OPT 2.7B	8, 16, 32	True / False	6
Qwen3.5	8, 16, 32	True / False	6
Total			18

For all configurations, an alpha of 17 and a dropout rate of 0.05 were used, and the vision encoder was frozen during training.

3.3 Experiments

Evaluation Metrics

Model performance is assessed using six standard captioning metrics: BLEU-1 through BLEU-4 [30], measuring n -gram overlap with reference captions; ROUGE-L [31], capturing sentence-level structural similarity; CIDEr [32], rewarding consensus across multiple human references; METEOR, assessing semantic similarity through synonym matching and stemming; and SPICE, evaluating scene graph-based semantic content. Higher values indicate better agreement with human-annotated captions across all metrics.

Implementation Details

ATHENA is implemented using PyTorch and the HuggingFace Transformers library. All images are resized to 224×224 pixels and processed through the enhancement pipeline before training and inference. All models are fine-tuned for 3 epochs using the 8-bit Paged AdamW optimizer with a learning rate of 2×10^{-4} , a cosine learning rate scheduler with 100 warmup steps, and an effective batch size of 16 (per-device batch size of 2 with 8 gradient accumulation steps). Fine-tuning and experiments are conducted on an NVIDIA A100 40 GB GPU using Flash Attention 2. The following custom system prompt is used during training: “*You are a VLM attached to an underwater robot. Describe the scene. Be concise but thorough.*” To assess deployment feasibility, all backbones are quantized to 4-bit NF4 format using the *bitsandbytes* library and served via a *FastAPI* server, and their VRAM footprints and inference latencies are recorded.

Chapter 4

Analysis and Discussion

This chapter presents the experimental results of the ATHENA pipeline across two datasets: the original underwater image captioning dataset by Li et al. [1] and the UW Bench dataset [2]. The image enhancement pipeline is applied to both datasets during training and evaluation. We first report the backbone and hyperparameter ablation results on the Li et al. dataset, then analyze the effect of the image enhancement pipeline across both datasets, with particular focus on the UW Bench dataset where images are substantially more degraded, and compare ATHENA against frontier vision-language models on the UW Bench dataset as a common zero-shot benchmark. Finally, we analyze computational efficiency and deployment feasibility.

4.1 Ablation Study Results

Following the ablation setup described in Section 3, we evaluate all LoRA rank and DoRA configurations on the original Li et al. dataset [1], whose images are relatively clear and captured under favorable conditions. Table 4.1 reports the best-performing configuration per backbone; full per-configuration results for all 18 configurations, including a visual comparison, are in Appendix C.

Table 4.1: Best configuration per backbone

Model	Config	B-1	B-2	B-3	B-4	R-L	C	S	M
SmolVLM	LoRA32 DoRA=Y	0.7820	0.6474	0.5327	0.4332	0.6574	1.2810	0.2610	0.3241
BLIP-2	LoRA32 DoRA=Y	0.6639	0.5317	0.4229	0.3335	0.5322	0.8740	0.2709	0.3335
Qwen3.5	LoRA8 DoRA=Y	0.7877	0.6535	0.5473	0.4568	0.6765	1.4464	0.2918	0.3572

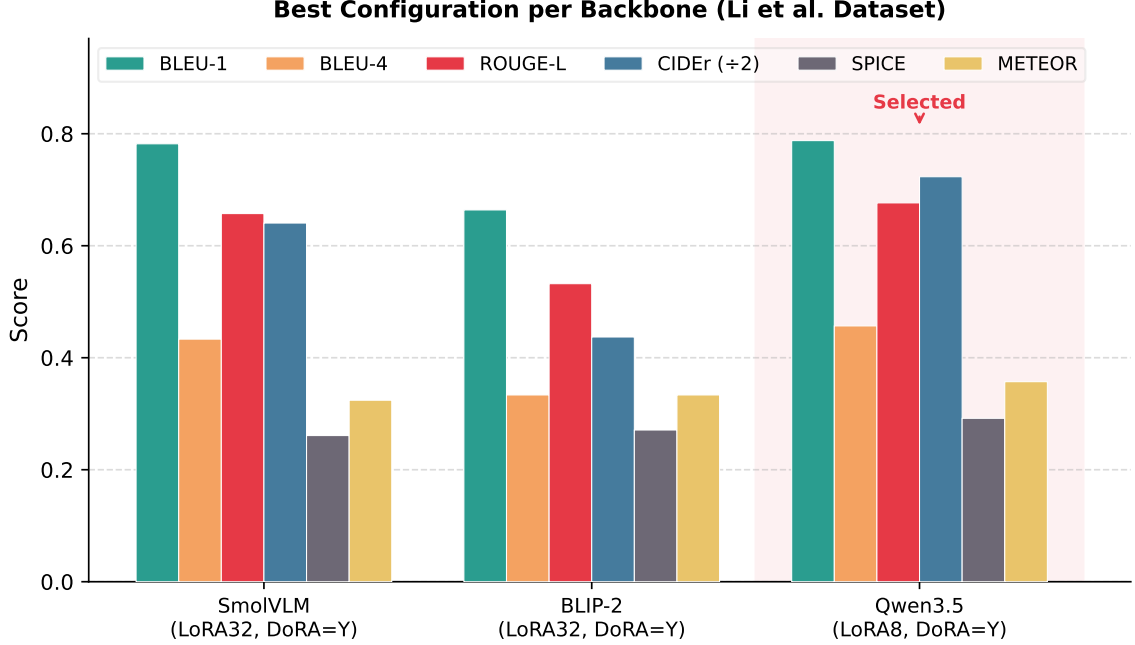


Figure 4.1: Best-configuration scores per backbone (visual representation of Table 4.1).

As shown in Figure 4.1, BLIP-2 showed the lowest overall performance across all configurations, with LoRA32 DoRA=Y as its best result, suggesting the model struggles to adapt to the underwater domain even with fine-tuning. SmolVLM converged effectively at LoRA32 DoRA=Y, achieving strong BLEU scores competitive with the larger Qwen3.5. Qwen3.5 performed most consistently and achieved the highest scores overall, with LoRA8 DoRA=Y as its best configuration, indicating its stronger pretrained representations benefit from weight decomposition even at low rank. Qwen3.5 is therefore selected as the ATHENA deployment backbone.

Effect of Image Preprocessing

To isolate the contribution of the image enhancement pipeline, Qwen3.5 was evaluated with and without preprocessing across all LoRA rank and DoRA configurations on both datasets. Table 4.2 shows the best result per condition on the original Li et al. dataset; full per-rank results are in Appendix C.

Table 4.2: Preprocessing ablation on the Li et al. dataset (Qwen3.5, best LoRA rank per condition)

DoRA	Prep	Best r	B-1	B-4	R-L	CIDEr	SPICE	METEOR
N	ON	16	0.7785	0.4419	0.6686	1.3928	0.2845	0.3471
N	OFF	8	0.7809	0.4474	0.6740	1.4234	0.2790	0.3485
Y	ON	8	0.7796	0.4449	0.6721	1.4457	0.2883	0.3503
Y	OFF	8	0.7877	0.4568	0.6765	1.4464	0.2918	0.3572

Across both DoRA settings, preprocessing provides no meaningful improvement on the Li et al. dataset; the Prep=OFF configurations match or slightly outperform Prep=ON on every metric. This is consistent with the nature of that dataset: its images are relatively clear, with limited turbidity or color distortion, leaving little room for enhancement to contribute. The preprocessing pipeline is therefore not necessary when input images are already of adequate visual quality.

To assess the pipeline’s contribution under realistic degraded conditions, we re-evaluated Qwen3.5 on the UWBench dataset [2], which features substantially more challenging images with greater occlusion, turbidity, color distortion, and variable illumination, with and without preprocessing across all LoRA rank and DoRA configurations. All scores are reported as percentages ($\times 100$). Table 4.3 shows the best result per condition; full per-rank results are in Appendix C.

Enabling preprocessing consistently improves performance across both DoRA settings. With DoRA disabled, the best preprocessing-on configuration (LoRA32) gains

Table 4.3: Preprocessing ablation on the UWBench dataset (Qwen3.5, best LoRA rank per condition)

DoRA	Prep	Best r	B-1	B-4	R-L	CIDEr	SPICE	METEOR
N	ON	32	48.30	16.21	36.21	61.10	34.22	40.02
N	OFF	16	44.95	14.61	33.73	54.76	31.72	37.25
Y	ON	8	48.83	16.14	35.94	59.16	34.17	40.27
Y	OFF	16	35.55	11.32	26.15	38.41	24.80	29.24

3.35 BLEU-1 and 6.34 CIDEr points over the best preprocessing-off configuration. The gap is substantially larger with DoRA enabled: preprocessing-on (LoRA8) leads by 13.28 BLEU-1 points and 20.75 CIDEr points over preprocessing-off (LoRA16). This amplified sensitivity suggests that weight decomposition makes the model more responsive to input quality, since the directional and magnitude updates of DoRA more closely mirror full fine-tuning and therefore depend more on clean, enhanced inputs. METEOR and SPICE follow the same trend across both DoRA settings.

Taken together with the marginal impact observed on the original dataset, these results support a key finding: **the image enhancement pipeline is not universally necessary, but becomes essential when input images are visually degraded**. In the original Li et al. dataset, where images are already clear, the model can extract sufficient visual features without enhancement. In the UWBench dataset, and by extension in real constrained underwater deployments where turbidity, occlusion, and color distortion are common, the pipeline provides a measurable performance benefit. Although DoRA=True exhibits greater sensitivity to preprocessing quality, the DoRA=False configuration with LoRA rank 32 and preprocessing enabled achieves superior performance on four out of six primary metrics on UWBench (B-4, ROUGE-L, CIDEr, and SPICE). The configuration of Qwen3.5 with preprocessing enabled, DoRA=False, and LoRA rank 32 is therefore adopted as the

ATHENA deployment configuration. Figure 4.2 illustrates this contrast directly: the top panel shows Preprocessing ON and OFF scores on the Li et al. dataset, where bars are nearly identical across all metrics; the bottom panel shows the same comparison on UW Bench, where Preprocessing ON consistently and clearly outperforms Preprocessing OFF.



Figure 4.2: Preprocessing ON vs. OFF across both datasets (Qwen3.5, DoRA=False).

4.2 Benchmark Comparison

On the original Li et al. dataset [1], ATHENA is compared against a range of established image captioning methods, including CNN-RNN models (Bi-LSTM [33]), attention-based models (Up-Down [34]), and Transformer-based approaches (S2 Transformer [35]). This comparison uses the same dataset all models were trained on, providing a direct measure of ATHENA’s advantage over prior specialized captioning architectures. Results are shown in Table 4.4, where B-N, M, R-L, C, and S denote BLEU-N, METEOR, ROUGE-L, CIDEr, and SPICE, respectively; best results are in **bold** and second-best are underlined.

Table 4.4: Benchmark comparison on the Li et al. dataset [1]

Model	B-1 ↑	B-2 ↑	B-3 ↑	B-4 ↑	M ↑	R-L ↑	C ↑	S ↑
Bi-LSTM [33]	0.6813	0.5554	0.4534	0.3661	0.2761	0.5661	1.0747	0.2163
Up-Down [34]	0.7064	0.5807	0.4653	0.3585	0.2775	0.5615	1.1680	0.2114
S2 Trans. [35]	0.7222	0.5958	0.4877	0.3938	<u>0.3028</u>	<u>0.5944</u>	1.2481	0.2311
UICM-SOFF [1]	<u>0.7249</u>	<u>0.5980</u>	<u>0.4920</u>	<u>0.3953</u>	0.2988	0.5888	<u>1.2831</u>	<u>0.2368</u>
ATHENA (Ours)	0.7877	0.6535	0.5473	0.4568	0.3572	0.6765	1.4464	0.2918

ATHENA outperforms all compared models across all eight metrics. The reported ATHENA configuration uses Qwen3.5 with LoRA rank 8 and DoRA enabled, with preprocessing disabled, as the image enhancement pipeline was found to have only marginal impact on this dataset’s relatively clear images and the best scores are achieved without it. Prior model results are taken from Li et al. [1], and ATHENA’s scores are appended in the last row.

Comparison with Frontier Vision-Language Models

To further contextualize ATHENA’s performance, we evaluate it against a broad set of frontier VLMs in a zero-shot setting on the UWBench dataset [2], the more challenging of the two datasets described in Section 3. ATHENA uses the best configuration identified above: Qwen3.5 with preprocessing enabled, DoRA=False, and LoRA rank 32. All scores are reported as percentages ($\times 100$). Frontier model results are reproduced from Zhang et al. [2], with ATHENA appended as the last row. Results are shown in Table 4.5, where B-N, M, R-L, C, and S denote BLEU-N, METEOR, ROUGE-L, CIDEr, and SPICE, respectively; best results are in **bold** and second-best are underlined.

Table 4.5: Benchmark comparison on the UWBench dataset [2]

Method	B-1 $\uparrow$	B-2 $\uparrow$	B-3 $\uparrow$	B-4 $\uparrow$	M $\uparrow$	R-L $\uparrow$	C $\uparrow$	S $\uparrow$
ATHENA (Ours)	<u>48.30</u>	31.79	22.13	16.21	40.02	36.21	<u>61.10</u>	<u>34.22</u>
GPT-5	49.17	<u>31.40</u>	<u>21.14</u>	<u>14.90</u>	<u>27.08</u>	<u>34.54</u>	66.10	35.41
GPT-5-mini	43.41	23.86	13.60	8.11	23.59	26.16	36.53	26.58
GPT-4o	40.24	22.51	12.95	7.97	22.98	26.79	33.10	28.73
GLM-4.5V-106B	41.96	23.41	13.67	8.41	22.45	25.46	31.48	26.68
Qwen3-VL-30B-Instruct	41.07	22.90	13.35	8.24	23.37	25.18	31.61	27.24
Qwen3-VL-30B-Thinking	40.82	22.64	13.28	8.19	21.74	24.86	29.22	25.79
Qwen2.5-VL-72B	40.07	22.09	12.77	7.90	21.15	25.14	31.24	27.40
GLM-4.1V-9B	40.28	22.21	12.87	7.87	21.15	25.02	26.40	23.74
InternVL-3.5-38B	36.94	20.21	11.58	7.14	20.48	24.38	26.38	26.00
InternVL-3.5-241B	35.88	19.87	11.50	7.13	20.23	24.20	23.60	26.18
Gemini-2.5-Flash	33.87	16.90	8.72	4.83	23.01	22.53	10.21	24.53
Qwen2.5-VL-7B	32.60	17.31	9.83	6.06	18.17	22.85	11.87	24.72
InternVL-3.5-1B	31.73	17.68	10.40	6.57	17.89	22.92	11.16	24.44
Qwen2.5-VL-3B	30.26	16.24	9.43	5.91	17.34	22.84	8.53	24.04

ATHENA ranks first on BLEU-2 through BLEU-4, METEOR, and ROUGE-L, and second on BLEU-1, CIDEr, and SPICE, with only GPT-5 exceeding it on those metrics. ATHENA’s METEOR of 40.02 substantially exceeds GPT-5’s 27.08 and all other models, indicating stronger lexical alignment with human reference captions;

this is particularly significant for a 1.37B-parameter model under 4-bit NF4 quantization competing against models up to 100B parameters evaluated zero-shot. GPT-5 leads on CIDEr and SPICE, reflecting stronger consensus-based semantic coverage, while ATHENA’s domain-specific fine-tuning drives superior n -gram precision and recall.

Figure 4.3 further illustrates this performance profile by normalizing each metric to the observed maximum and plotting ATHENA alongside GPT-5, GPT-5-mini, and GPT-4o, the top three frontier models by semantic captioning metrics (CIDEr, SPICE, METEOR, and ROUGE-L). ATHENA leads on METEOR and ROUGE-L, while GPT-5 leads on CIDEr and SPICE, reflecting a trade-off between lexical fluency and consensus-based semantic coverage.

Metric Profile: ATHENA vs. Top Frontier Models on UWBench
(Each axis scaled to best observed score per metric)

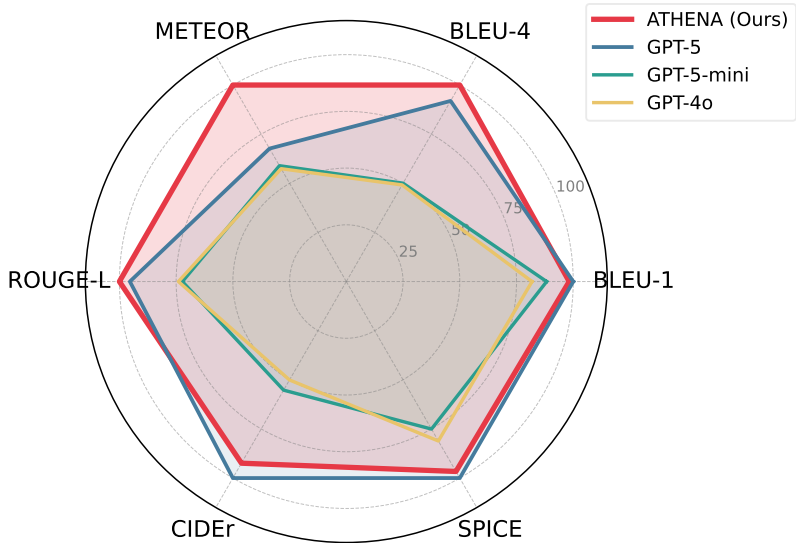


Figure 4.3: ATHENA vs. Top 3 Frontier VLMs on UWBench.

This performance gain reflects ATHENA’s vision-language design and its capacity

to address low contrast, occlusions, and complex underwater scene structures. Figures 4.4 and 4.5 illustrate the effect of the seven-step enhancement pipeline on sample images from the Li et al. dataset and UWBench, respectively. The enhancement effect is less pronounced on the Li et al. images, which exhibit milder visual degradation, while improvements are more visually apparent on UWBench images, consistent with the preprocessing ablation findings.

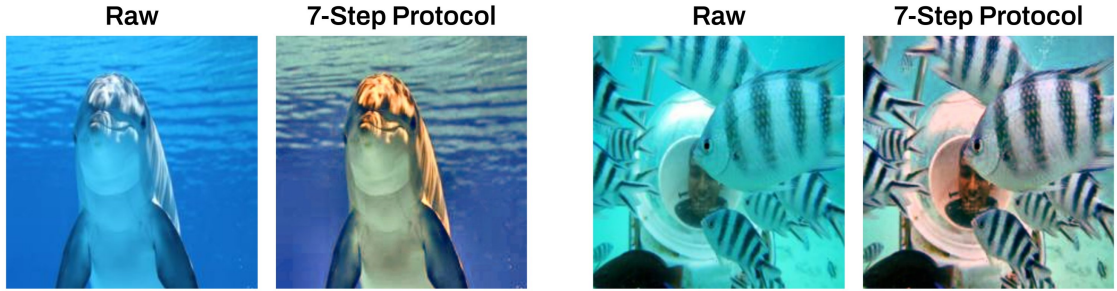


Figure 4.4: Sample raw and preprocessed image pairs from the Li et al. dataset

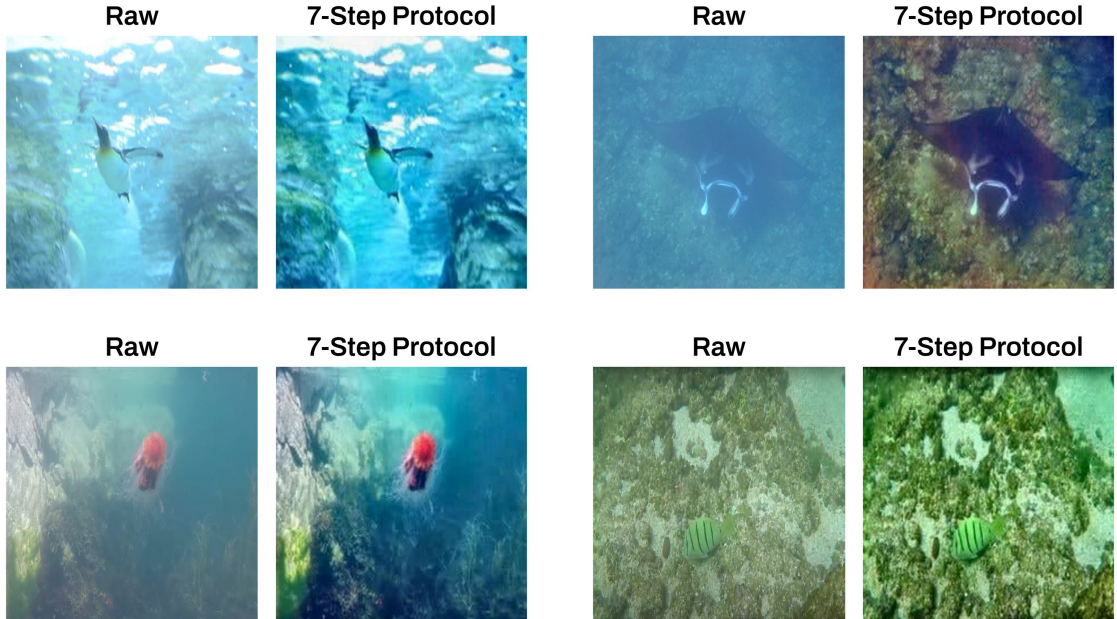


Figure 4.5: Sample raw and preprocessed image pairs from the UWBench dataset

4.3 Computational Efficiency and Deployment Analysis

To validate the feasibility of ATHENA’s deployment on resource-constrained AUV hardware, we evaluated the computational footprint and inference latency of all three backbones under 4-bit NF4 quantization using the *bitsandbytes* library, deployed as a *FastAPI* server. Table 4.6 summarizes the resource utilization.

Table 4.6: Resource utilization under 4-bit NF4 quantization

Model	Parameters	VRAM (fp16)	VRAM (Quantized)
SmolVLM	1,267,694,576	1.49 GB	604.48 MB
Qwen3	1,238,994,944	1.50 GB	590.80 MB
BLIP-2	2,025,179,136	2.20 GB	965.68 MB
Qwen3.5	1,369,148,224	1.77 GB	652.86 MB

Although Qwen3.5 has a slightly larger parameter count (1.37B) than SmolVLM (1.27B), its 4-bit quantized VRAM footprint of only 653 MB is comparable to SmolVLM’s 604 MB, and both are well within the constraints of standard edge computing modules. The NVIDIA Jetson Orin series typically offers 8 GB of unified memory, leaving substantial headroom for system overhead and local image storage. BLIP-2, by contrast, requires 966 MB under 4-bit quantization, making it the least efficient option despite its intermediate parameter count. Critically, Qwen3.5 achieves an inference latency of approximately 30 seconds per caption. The image preprocessing pipeline adds only 0.03 seconds per frame across all backbones. These latencies are well within acceptable bounds for underwater acoustic communication scenarios, where data transmission rates rather than onboard processing are the primary bottleneck.

Chapter 5

Conclusion

This chapter summarizes the contributions of this work, reflects on the key findings across all experiments, and identifies directions for future research.

5.1 Conclusion

This thesis presented ATHENA, an image captioning pipeline designed for constrained underwater environments where visual degradation, limited onboard compute, and low-bandwidth acoustic communication impose strict practical constraints. ATHENA integrates a seven-step image enhancement pipeline with a parameter-efficiently fine-tuned vision-language model to generate structured scene descriptions that can be transmitted over acoustic channels to a remote operator.

Through systematic ablation across three VLM backbones (SmolVLM-Instruct, BLIP-2 OPT 2.7B, and Qwen3.5-VL), six LoRA rank and DoRA configurations, and two datasets of varying visual complexity, this work produced three central findings.

First, **Qwen3.5-VL with LoRA rank 32 and DoRA disabled** is the optimal backbone configuration for deployment. Despite having only 1.37 billion parameters, it achieves the highest scores across the majority of captioning metrics on UW Bench,

and under 4-bit NF4 quantization occupies only 653 MB of VRAM, which fits comfortably within the memory budget of embedded platforms such as the NVIDIA Jetson Orin series. Its inference latency of under 30 seconds per caption is acceptable for underwater acoustic communication scenarios where transmission bandwidth, not onboard computation, is the primary bottleneck.

Second, **the image enhancement pipeline is conditionally necessary**. On the original Li et al. dataset [1], whose images are relatively clear and captured under favorable conditions, preprocessing produced only marginal score improvements, suggesting the VLM can extract sufficient visual features without enhancement when image quality is adequate. On UW Bench [2], which presents substantially more challenging images with occlusion, turbidity, and color distortion characteristic of real constrained deployments, enabling preprocessing yielded gains of 3.4 BLEU-1 points and 6.3 CIDEr points for the deployment configuration (DoRA=False, LoRA32 vs. best off configuration), with substantially larger gains observed under DoRA=True settings. This confirms that the pipeline’s value is tied to the severity of visual degradation in the deployment environment.

Third, **ATHENA is competitive with frontier VLMs at a fraction of their scale**. Evaluated zero-shot on UW Bench against 14 frontier models ranging from 3B to over 100B parameters, ATHENA ranks first on BLEU-2 through BLEU-4, METEOR, and ROUGE-L, and second overall behind only GPT-5. Its METEOR score of 40.02 is substantially higher than all other evaluated models including GPT-5 (27.08), indicating that domain-specific fine-tuning on underwater imagery produces strong lexical alignment with human reference captions that generalist models at any scale cannot easily replicate zero-shot.

Together, these results demonstrate that a carefully fine-tuned compact VLM paired with an appropriate image enhancement strategy can achieve state-of-the-art underwater scene captioning while remaining deployable on resource-constrained autonomous platforms.

5.2 Summary of Contributions

This work presents ATHENA, an end-to-end image captioning pipeline integrating a seven-step underwater image enhancement protocol with a parameter-efficiently fine-tuned VLM, designed for low-bandwidth deployment on autonomous underwater vehicles. A systematic ablation across three VLM backbones and six LoRA rank and DoRA configurations establishes Qwen3.5-VL (LoRA32, DoRA=False) as the optimal deployment configuration, with controlled experiments across two datasets showing that the image enhancement pipeline is conditionally necessary: beneficial for degraded inputs but unnecessary when images are already clear.

ATHENA is further benchmarked against 14 frontier VLMs on UW Bench in a zero-shot setting, demonstrating that a fine-tuned 1.37B-parameter model outperforms much larger systems on most metrics and establishing a strong baseline for future underwater captioning work. Deployment feasibility is confirmed through quantitative measurement of VRAM footprint, inference latency, and preprocessing overhead under 4-bit NF4 quantization, showing that ATHENA meets the resource constraints of standard edge computing hardware used in underwater robotics.

5.3 Future Research

Several directions remain open for extending this work.

Adaptive preprocessing. The current pipeline applies a fixed seven-step enhancement sequence to every image regardless of its actual degradation level. An adaptive controller trained to select or weight enhancement steps based on estimated water quality, turbidity, and visibility could reduce unnecessary processing on clear images while improving output quality on degraded ones.

Real-time onboard evaluation. All experiments in this work were conducted on a server-grade NVIDIA A100 GPU. Validating ATHENA’s latency and output quality on actual edge hardware (e.g., NVIDIA Jetson Orin NX) under realistic acoustic communication constraints would constitute a critical step toward operational deployment.

Multi-task extension. UW Bench supports object referring and visual question answering alongside image captioning. Extending ATHENA to handle these tasks through multi-task fine-tuning could increase its utility as a general onboard underwater perception module, supporting not only scene narration but also targeted object localization and interactive querying by remote operators.

Appendix A

Environment and Source Codes

A.1 Environment

All fine-tuning and evaluation experiments were conducted on a single NVIDIA A100 40 GB SXM4 GPU. The software environment is listed in Table A.1.

Table A.1: Training and evaluation environment	
Component	Version / Specification
GPU	NVIDIA A100 40 GB SXM4
CUDA	12.1
Python	3.11
PyTorch	2.2.0
Transformers (HuggingFace)	4.40.0
PEFT	0.10.0
bitsandbytes	0.43.0
Flash Attention	2.5.6
Accelerate	0.29.0

Quantization for deployment feasibility testing used the *bitsandbytes* library with 4-bit NF4 format and double quantization enabled. Inference was served via a *FastAPI* server to measure end-to-end latency, including image preprocessing and model generation time.

A.2 Source Codes

The implementation covers five main components: (1) the seven-step image enhancement pipeline, (2) dataset loading and caption formatting, (3) LoRA/DoRA fine-tuning via HuggingFace PEFT, (4) metric evaluation using the *pycocoevalcap* library for BLEU, ROUGE-L, CIDEr, METEOR, and SPICE, and (5) the FastAPI inference server with 4-bit NF4 quantization.

Fine-tuning used the *SFTTrainer* from TRL with the following training arguments common to all configurations: 3 training epochs, 8-bit paged AdamW optimizer, learning rate 2×10^{-4} , cosine learning rate scheduler with 100 warmup steps, per-device batch size of 2 with 8 gradient accumulation steps (effective batch size 16), and Flash Attention 2. The vision encoder was frozen in all experiments. All modules apply the same custom system prompt during training: *“You are a VLM attached to an underwater robot. Describe the scene. Be concise but thorough.”*

Appendix B

Datasets

B.1 Li et al. Underwater Image Captioning Dataset

The primary training and evaluation dataset is the underwater image captioning dataset introduced by Li et al. [1]. It contains 3,176 high-resolution underwater images across ten scene categories, with fish (1,861 images), coral reefs (1,587 images), and divers (850 images) being the most frequent. Each image is annotated with five distinct human-written captions contributed by volunteers with marine domain expertise, yielding 15,880 captions in total. Captions describe object morphology, color, quantity, motion, and scene context, providing vocabulary and syntactic diversity for training. Images were resized to 224×224 pixels for all experiments.

Table B.1: Li et al. dataset statistics	
Statistic	Value
Total images	3,176
Captions per image	5
Total captions	15,880
Scene categories	10
Image resolution	224×224 (resized)

B.2 UWBench Dataset

UWBench [2] is used as an extended evaluation benchmark to assess ATHENA under more visually challenging conditions. Figure B.1 reproduces the construction overview from Zhang et al. [2].

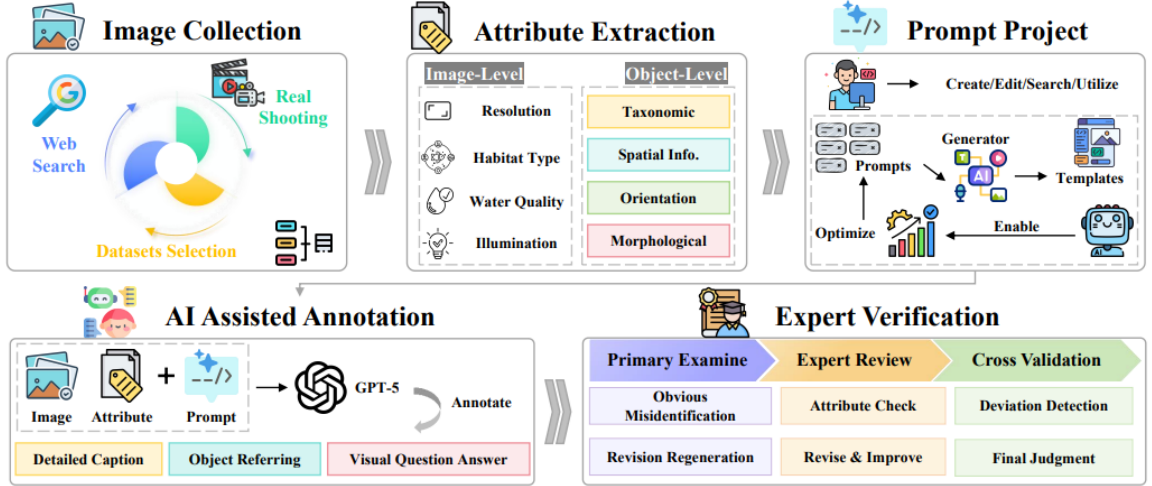


Figure B.1: UWBench construction pipeline, reproduced from Zhang et al. [2].

Image collection. Images were gathered from web searches, existing underwater detection datasets, and in-situ robotic footage. From over 35,000 candidates, 15,003 were selected after filtering for quality, duplicates, and diversity across habitat type, water conditions, and scene complexity.

Attribute extraction. Each image was annotated with image-level attributes (habitat, water quality, illumination) and object-level attributes (taxonomy, bounding box, spatial relationships, morphology), which served as structured inputs to the annotation pipeline.

AI-assisted annotation. GPT-5 generated initial captions, object referring expressions, and visual QA pairs for each image using task-specific prompts grounded

in the extracted attributes.

Expert verification. All outputs underwent three-stage review by marine biology experts: error checking by trained annotators, validation of taxonomic accuracy by senior biologists, and cross-validation of 2,000 images for inter-annotator agreement, totaling over 600 hours.

The resulting dataset contains 15,003 expert-verified captions, 15,281 object referring expressions, and 124,983 visual QA pairs. UWBench-Cap is partitioned into 10,454 training images and 4,549 test images.

Table B.2: UWBench statistics

Statistic	Value
Total images	15,003
Object categories	158
Captions (avg. words)	68.6
Object referring expressions	15,281
VQA pairs	124,983
Training split (Cap)	10,454 images
Test split (Cap)	4,549 images

Appendix C

Full Experimental Results

C.1 Backbone Hyperparameter Ablation

Tables C.1 through C.3 report all 18 configurations from the backbone ablation on the Li et al. dataset. B-N, R-L, C, S, and M denote BLEU-N, ROUGE-L, CIDEr, SPICE, and METEOR; all metrics are on a 0–1 scale. Bold indicates the best result per metric within each backbone. Figure C.1 visualizes scores across all configurations simultaneously, grouped by backbone.

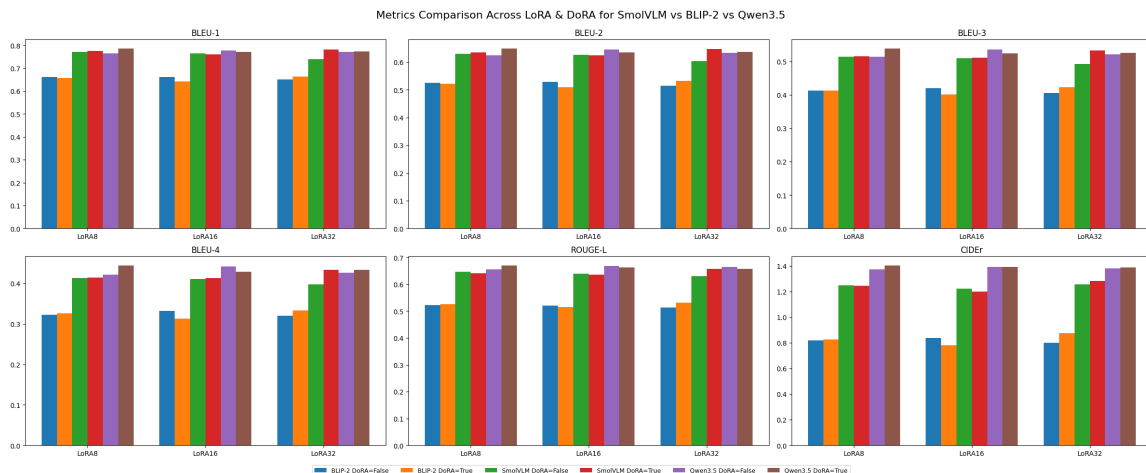


Figure C.1: Metrics comparison across all 18 configurations, grouped by backbone.

Table C.1: SmolVLM-Instruct: all configurations

Model	Config	B-1	B-2	B-3	B-4	R-L	C	S	M
SmolVLM	LoRA8 DoRA=N	0.7710	0.6300	0.5169	0.4137	0.6466	1.2475	0.2675	0.3301
SmolVLM	LoRA8 DoRA=Y	0.7760	0.6339	0.5160	0.4148	0.6419	1.2449	0.2723	0.3325
SmolVLM	LoRA16 DoRA=N	0.7651	0.6256	0.5106	0.4103	0.6398	1.2234	0.2616	0.3247
SmolVLM	LoRA16 DoRA=Y	0.7611	0.6240	0.5116	0.4126	0.6358	1.1988	0.2639	0.3247
SmolVLM	LoRA32 DoRA=N	0.7398	0.6027	0.4931	0.3979	0.6307	1.2563	0.2569	0.3236
SmolVLM	LoRA32 DoRA=Y	0.7820	0.6474	0.5327	0.4332	0.6574	1.2810	0.2610	0.3241

Table C.2: BLIP-2 OPT 2.7B: all configurations

Model	Config	B-1	B-2	B-3	B-4	R-L	C	S	M
BLIP-2	LoRA8 DoRA=N	0.6612	0.5247	0.4124	0.3220	0.5218	0.8209	0.2656	0.3230
BLIP-2	LoRA8 DoRA=Y	0.6564	0.5217	0.4134	0.3257	0.5252	0.8206	0.2793	0.3328
BLIP-2	LoRA16 DoRA=N	0.6610	0.5275	0.4198	0.3318	0.5209	0.8387	0.2698	0.3279
BLIP-2	LoRA16 DoRA=Y	0.6434	0.5093	0.4009	0.3131	0.5146	0.7798	0.2608	0.3196
BLIP-2	LoRA32 DoRA=N	0.6498	0.5142	0.4053	0.3198	0.5140	0.7996	0.2668	0.3232
BLIP-2	LoRA32 DoRA=Y	0.6639	0.5317	0.4229	0.3335	0.5322	0.8740	0.2709	0.3335

Table C.3: Qwen3.5-VL: all configurations

Model	Config	B-1	B-2	B-3	B-4	R-L	C	S	M
Qwen3.5	LoRA8 DoRA=N	0.7809	0.6467	0.5390	0.4474	0.6740	1.4234	0.2790	0.3485
Qwen3.5	LoRA8 DoRA=Y	0.7877	0.6535	0.5473	0.4568	0.6765	1.4464	0.2918	0.3572
Qwen3.5	LoRA16 DoRA=N	0.7715	0.6401	0.5334	0.4422	0.6615	1.4340	0.2830	0.3502
Qwen3.5	LoRA16 DoRA=Y	0.7778	0.6451	0.5397	0.4502	0.6680	1.4420	0.2868	0.3476
Qwen3.5	LoRA32 DoRA=N	0.7749	0.6300	0.5169	0.4237	0.6604	1.3648	0.2744	0.3363
Qwen3.5	LoRA32 DoRA=Y	0.7813	0.6374	0.5236	0.4271	0.6624	1.3705	0.2764	0.3381

C.2 Preprocessing Ablation on UWBench

Tables C.4 and C.5 show the full per-rank results for the preprocessing ablation on UWBench (Qwen3.5). All scores are percentages ($\times 100$); bold indicates the best per group.

Table C.4: Preprocessing ablation on UWBench: DoRA disabled, all LoRA ranks

Prep	r	B-1	B-2	B-3	B-4	R-L	C	SPICE	METEOR
ON	8	48.76	31.87	22.05	16.06	35.88	58.90	34.21	40.22
ON	16	47.46	31.27	21.83	16.00	36.16	60.57	33.89	39.44
ON	32	48.30	31.79	22.13	16.21	36.21	61.10	34.22	40.02
OFF	8	42.97	27.64	18.75	13.38	30.97	50.35	29.32	34.96
OFF	16	44.95	29.32	20.19	14.61	33.73	54.76	31.72	37.25
OFF	32	41.95	27.33	18.76	13.52	31.89	49.92	29.99	35.02

Table C.5: Preprocessing ablation on UWBench: DoRA enabled, all LoRA ranks

Prep	r	B-1	B-2	B-3	B-4	R-L	C	SPICE	METEOR
ON	8	48.83	31.95	22.12	16.14	35.94	59.16	34.17	40.27
ON	16	47.25	31.08	21.67	15.88	35.93	59.44	33.79	39.24
ON	32	48.51	31.85	22.17	16.24	36.18	60.05	34.30	40.13
OFF	8	34.33	22.18	15.14	10.89	25.93	43.21	24.54	29.26
OFF	16	35.55	22.98	15.72	11.32	26.15	38.41	24.80	29.24
OFF	32	33.94	21.96	15.03	10.83	25.45	38.91	24.37	28.81

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